MIL-W-13169B
AUGUST 1, 1959
SUPERSEDING
MIL-W-13169A
29 AUGUST 1957

MILITARY SPECIFICATION

WIRE, ELECTRICAL (FOR INSTRUMENT TEST LEADS)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 Scope. This specification covers single-conductor, AWG size 18, stranded, electrical wire, insulated for voltages up and including 1,000 volts root-mean-square (rms) with Buna-S compound. It is primarily intended for use as instrument test leads.

1.2 Classification. Wire covered by this specification shall be type TL-G, having an insulation of one of the following colors, black, red, or green, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

J-C-98 — Cable and Wire, Insulated; Methods of Sampling and Testing.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-C-572 — Cords, Yarns and Monofilaments — Organic Synthetic Fiber.

MIL-W-3861 — Wire, Electrical (Bare Copper).

MIL-I-3930 — Insulating and Jacketing Compounds, Electrical (For Cable, Cord and Wire).

MIL-C-12000 — Cable, Cord, and Wire, Electric, Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-104 — Limits for Electrical Insulation Color.

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Copper strands.* The strands comprising the conductor shall be tin-coated, soft or drawn-and-annealed copper wire conforming to Specification MIL-W-3861.

3.1.2 *Separators.* A separator may be used to provide free stripping of the insulation.

FSC 6145

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When a separator is used, it shall completely cover the conductor. It shall be type AR, form Y, (acetate-rayon thread) or type P (Polyamide) conforming to Specification MIL-C-572. The separator may be omitted if a free-stripping insulation is used.

3.1.3 *Insulation compound.* The insulation compound shall be Buna-S, made with such fillers, antioxidants, organic accelerators, and curing agents as are necessary to provide a compound having long life in service and in storage. The compound used shall contain not more than 0.5 percent of free sulfur. The material shall be homogeneous, tough, and elastic, and shall be properly vulcanized (cured) to conform to the requirements of this specification. It shall be free of blisters, wrinkles, cracks, or other defects that may have a deleterious effect on the finished wire.

3.1.3.1 *Sunproofing wax.* For each 100 pounds of Buna-S polymer, a minimum of 4 pounds of acceptable sunproofing wax shall be used (see 6.3).

3.1.3.2 *Color.* The shade of color (see 1.2) shall fall within the light and dark limits of Standard MIL-STD-104.

3.2 Construction.

3.2.1 *Conductor.* The conductor shall be AWG size 18, type B, class 0 (for severe flexing service) in accordance with Specification MIL-W-3861.

3.2.2 *Insulation thickness.* The insulation applied over the conductor shall be well-centered, and shall have a thickness of not less than 40 mils of Buna-S compound (see 3.1.3 and 4.5.1).

3.2.3 *Outside diameter.* The outside diameter of the finished wire shall not exceed 150 mils, when measured as specified in 4.5.1.

3.3 Physical properties of insulation. The vulcanized Buna-S insulation shall conform to the physical-property requirements specified in table I (see 4.5.2).

TABLE I. *Physical-property requirements of insulation*

Physical property	Buna-S insulation
Unaged:	
Tensile strength, minimum psi	1,000
Elongation, minimum inches	8
Set, maximum, inch	½
After 94 hours oxygen-bomb at 70° C.:	
Tensile strength, minimum percent of original.	75
Elongation, minimum, inches	6
Cold bend	(1)
Cold-tension recovery, minimum, percent	20

(1) There shall be no cracking of insulation.

3.4 Performance.

3.4.1 *Spark.* The finished wire shall withstand a spark-test potential of not less than 8,000 volts rms, when tested as specified in 4.5.6. A certification to the effect that the spark test has been performed shall accompany the wire when it is submitted for acceptance inspection (see 4.5.1).

3.4.2 *Dielectric strength.* The finished wire shall withstand without breakdown the test specified in 4.5.3.

3.4.3 *Insulation resistance.* The insulation resistance of the finished wire shall be not less than 1,000 megohms — 1,000 feet, when tested as specified in 4.5.4.

3.4.4 *Continuity.* The finished wire shall be continuous, when tested as specified in 4.5.5.

3.5 Workmanship. The wire shall be manufactured in accordance with paragraphs 3.2.1 thru 3.2.3 of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the sup-

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plier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Acceptance inspection of wire (see 4.4.1).

(b) Acceptance inspection of preparation for delivery (see 4.4.2).

4.3 Inspection, general.

4.3.1 *Definitions.*

4.3.1.1 *Inspection lot.* An inspection lot shall be all wires of any one color produced under substantially the same conditions and offered for inspection at one time.

4.3.1.2 *Unit of product.* The unit of product shall be the continuous length of wire contained on a spool or reel, or in a package.

4.3.1.3 *Sample.* The sample shall be defined as the number of specimens to be subjected to test.

4.3.1.4 *Specimen.* The specimen shall be the unit of product selected in an unbiased manner, which is subject to a test.

4.3.1.5 *Finished wire.* The finished wires are considered those wires in which all manufacturing operations have been completed and which are ready to be submitted for inspection.

4.4 Acceptance inspection.

4.4.1 *Acceptance inspection of wire.* Acceptance inspection shall consist of groups A, B and C inspection.

4.4.1.1 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected, using tightened inspection.

4.4.1.2 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table II. Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105.

4.4.1.2.1 When Standard MIL-STD-105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government, or by the contractor under supervision of the Government inspector with final responsibility for acceptance of the item resting with the Government inspector.

4.4.1.2.2 *Major and minor defects.* Major and minor defects shall be as defined in Standard MIL-STD-105 and as classified below:

Major defect

Broken strands in conductor.
Insulation thickness.
Nicked or torn insulation.
Occlusions in insulation.
Outside diameter.

Minor defect

Missing strands in conductor.
Free stripping.
Completeness of coverage of separator.
Color of insulation.
Rough surface of insulation.
Centering of insulation.

4.4.1.3 *Group B inspection.* Group B inspection shall consist of the examinations and tests specified in table III.

4.4.1.3.1 *Sampling procedure.* The sampling procedure shall be in accordance with the appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL shall be 6.5 percent and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection.

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TABLE II. *Group A inspection*

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and mechanical examination		4.5.1	1.0	4.0
Materials				
Copper strands	3.1.1			
Separators	3.1.2			
Insulation compound	3.1.3			
Color	3.1.3.2			
Construction				
Conductor	3.2.1			
Insulation thickness	3.2.2			
Outside diameter	3.2.3			
Workmanship	3.5			
Performance			1.0	(*)
Dielectric strength	3.4.2	4.5.3		
Insulation resistance	3.4.3	4.5.4		
Continuity	3.4.4	4.5.5		

* All electrical tests are major.

TABLE III. *Group B inspection*

Inspection or test	Requirement paragraph	Inspection paragraph	AQL
Unaged:			
Tensile strength	3.3	4.5.2	6.5%
Elongation	3.3	4.5.2	
Set	3.3	4.5.2	

4.4.1.3.2 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105.

4.4.1.4 *Group C inspection.* Group C inspection shall consist of the tests specified in table IV.

4.4.1.4.1 *Noncompliance.* If a sample fails to pass group C inspection, the manufacturer shall take corrective action on the process and on all units of product which can be corrected and which were manufactured with the same conditions, materials, processes, etc. and are considered subject to the same test failures. Acceptance inspection shall be discontinued until corrective action has been taken. After corrective action, sample units shall be subjected to the necessary group C inspection (all inspections, or failed inspections, at the option of the Government) group A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C inspection has shown that the corrective action was successful.

TABLE IV. *Group C inspection*

Inspection or test	Requirement paragraph	Inspection paragraph	Sampling plan
Aged (oxygen-bomb):			
Tensile strength	3.3	4.5.2	(*)
Elongation	3.3	4.5.2	(*)
Low temperature:			
Cold bend	3.3	4.5.2.1	(*)
Cold-tension recovery	3.3	4.5.2	(**)

* 6 specimens from the first lot presented for acceptance inspection and then 6 specimens from each 100,000 feet produced.

** 3 specimens from the first lot presented for acceptance inspection and then 3 specimens from each 100,000 feet produced.

4.4.2 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to table V and VII.

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TABLE V. *Classification of defects for preservation, packaging and marking.*

Major defects	Minor defects
1. Quantity per unit package not in accordance with the contract. 2. Item not properly cleaned prior to preservation (when applicable). 3. Incorrect packaging method applied. 4. Use of improper or defective material. 5. Cushioning or padding not applied or inadequate for the protection of the barrier material from projections, sharp edges or other similar features of the item. 6. Cushioning inadequate or not applied when required for the physical and/or mechanical protection of the item. 7. Stock number omitted, incorrect or illegible. 8. Nomenclature incorrect or missing. 9. Quantity of items in package not indicated.	1. Item not properly blocked or braced within the unit package to prevent movement. 2. Packaging material damaged. 3. Any item of marking information other than 7 thru 9 listed under major defects missing or illegible.

TABLE VI. *Classification of defects for packing and marking.*

Major defects	Minor defects
1. Box closure not in accordance with specification. 2. Type, grade, class and style of the shipping container do not meet the minimum requirements. 3. Strapping when required is omitted or inadequate, or incorrectly applied. 4. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement and/or damage. 5. Gross weight in excess of contract limitation. 6. Use of improper or defective material. 7. Shipping documents or packing list omitted (when required). 8. Stock number (when required) omitted or illegible. 9. Nomenclature (when required) omitted or illegible. 10. Quantity incorrect or missing. 11. Destination marking incorrect or illegible. 12. Service designation (color marking) missing. 13. Special marking and labeling requirements cited in the contract, such as MAP labels, shipment digit markings, etc, not applied. 14. Overseas code marking — illegible, incomplete or missing.	1. Unsealed carton. 2. Defective taping or sealing of carbon. 3. Any other box defect which may be considered minor by definition of MIL-STD-105. 4. Any item of required marking information other than 8 thru 14 listed under major defects, missing or illegible.

4.4.3 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in table V and VI and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL-STD-105. Inspection Level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the con-

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tract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be ten percent.

4.4.3.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.4.3.2 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the Contracting Officer for resolution.

4.4.3.3 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.5 Methods of examination and test.

4.5.1 *Visual and mechanical examination.* Wire shall be examined to verify that the materials, construction, and workmanship are in accordance with the applicable requirements (see 3.1, 3.2, and 3.5). The outside diameter of the finished wire (see 3.2.3) shall be measured by using the applicable portions of method 1011 of Specification J-C-98; thickness of insulation (see 3.2.2) by method 1018 of Specification J-C-98. The inspector shall also verify that the requirement for spark test is met (see 3.4.1).

4.5.2 *Physical properties of insulation (see 3.3).* The physical properties of insulation shall be tested in accordance with Specification MIL-I-3930 except that the low temperature cabinet test temperature shall be maintained within a range from — 38° C. to — 42° C.

4.5.2.1 *Cold bend.* One specimen of each color of finished wire shall be attached to a mandrel 0.188 inch in diameter. The specimen shall be suspended vertically, with the lower end weighted sufficiently to keep the specimen taut and to permit bending without handling. The mandrel and specimen shall be conditioned in the low-temperature cabinet for 7 days at — 40° ± 2° C. While at this temperature, the specimen shall be bent for five close turns around the mandrel at the rate of 15 ± 3 turns per minute. Upon completion of the test, the specimen shall be removed from the cabinet and the insulation shall be examined for cracks through a magnifying glass of at least 3-diameter magnification.

4.5.3 *Dielectric strength (see 3.4.2).* Wire shall be tested as specified in method 6111 of Specification J-C-98, except that —

(a) The test shall be performed on finished wire only.

(b) The test voltage shall be 4,000 volts rms, and shall be applied between the conductor and the water for a period of 60 $^{+5}_{-0}$ seconds from the time it is reached.

(c) The source of power shall be not less than 5-kilovolt-amperes capacity.

(d) The crest factor of the test voltage shall not differ by more than ± 10 percent from that of a sinusoidal wave when the transformer is loaded with the test specimen.

4.5.4 *Insulation resistance (see 3.4.3).* Insulation resistance of the finished wire shall be determined as specified in method 6031 of Specification J-C-98, except that —

(a) The insulation resistance shall be measured within 2 minutes after

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the dielectric-strength test (see 4.5.3).

(b) The test voltage shall be not less than 125 volts nor more than 500 volts dc.

(c) The polarity of the conductor shall be maintained negative with respect to the water.

(d) If the measurement is made at a temperature other than 15.6° C., the manufacturer shall correct the measured value of insulation resistance to the resistance at 15.6° C. If the insulation resistance is equal to or greater than that required by 3.4.3, when the measurement is made at a temperature greater than 15.6° C., no correction factor need be employed. The manufacturer shall demonstrate that the correction factor is accurate for his compound.

(e) The insulation-resistance test may be terminated in less than 1 minute if the galvanometer has ceased fluctuating and the reading indicates that a steady insulation-resistance value has been obtained.

4.5.5 *Continuity (see 3.4.4).* Continuity of the finished wire shall be determined by using a 10-volt power supply with a 10-volt lamp as an indicator.

4.5.6 *Spark test (see 3.4.1).* All finished wire shall be subjected to the spark test in accordance with method 6211 of Specification J-C-98.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Wire, electrical (for instrument test leads) shall be preserved and packaged in accordance with the applicable provisions of Specification MIL-C-12000.

5.1.2 *Level C.* Wire, electrical (for instrument test leads) shall be preserved and packaged in accordance with the applicable provisions of Specification MIL-C-12000.

5.2 Packing.

5.2.1 *Level A.* Wire, electrical (for instrument test leads) shall be packed for shipment as specified in Specification MIL-C-12000.

5.2.2 *Level B.* Wire, electrical (for instrument test leads) shall be packed for shipment as specified in Specification MIL-C-12000.

5.2.3 *Level C.* Wire, electrical (for instrument test leads) shall be packed for shipment as specified in Specification MIL-C-12000.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Specification MIL-C-12000.

6. NOTES

6.1 Intended use. Wire covered by this specification is primarily intended for use as instrument test leads.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Color required (see 1.2).

(c) Lengths required.

(d) Levels of packaging and packing, and applicable marking (see sec. 5).

6.2.1 *Indirect shipments.* The packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

6.3 Acceptable sunproofing wax. Information as to acceptable sunproofing wax may be obtained from the procuring activity.

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Custodians:

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Preparing activity:

Army—Signal Corps

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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